1. AWS Free Tier Setup :

Task	Steps
Create Account	- Go to https://aws.amazon.com/free- Sign up using a **non-primary email ID**.- Use a **debit/credit card** (Rs.2 verification charge).
Enable MFA for Root	- Login with root account.- Go to **IAM > Users > Security credentials**.- Under **Multi-Factor Authentication (MFA)**, select **Virtual MFA**.- Use **Google Authenticator** app to scan and link.
Create IAM User (Admin Access)	- Go to **IAM > Users > Add user**.- Username: `YourName_Dev`.- Enable **Programmatic access** and **AWS Management Console access**.- Attach existing policies directly → Choose `AdministratorAccess`.- Download `.csv` with credentials. Use this IAM user for daily tasks.

2.EC2 Hands-On Practice (Free Tier):

Task	Steps
AMI & Instance Type	- Go to **EC2 Console**.- Choose **Amazon Linux 2023** or **Ubuntu 22.04**.- Instance Type: `t2.micro` (Free Tier Eligible).
Key Pair	- Create a new key pair (RSA).- Download the `.pem` file (e.g., `aws_linux.pem`).
Security Group (Inbound Rules)	- Allow **SSH (Port 22)** for your IP.- Allow ICMP (for ping), optional.
Launch	- Select **Mumbai (ap-south-1)** or **Hyderabad (ap-south-2)** region.- Launch instance.
Connect via SSH	Open Terminal/CMD:`bash chmod 400 aws_linux.pem ssh -i "aws_linux.pem" ec2-user@<your-public-ip> `*(Ubuntu = `ubuntu@<ip>`)*

3.Launch a Windows EC2 Instance (t2.micro):

Task	Steps
AMI & Instance Type	- Choose **Microsoft Windows Server 2019 Base**.- Instance Type: `t2.micro`.
Key Pair & Launch	- Use existing or create new `.pem` key pair.- Launch in same region.
Security Group (Inbound Rules)	- Allow **RDP (Port 3389)**.
Connect via RDP	- After instance boots, choose **Get Windows Password**.- Upload `.pem`, decrypt to get username & password.- Use **Remote Desktop Connection (mstsc)**:` Computer: <your-public-ip> Username: Administrator Password: <decrypted-password> `

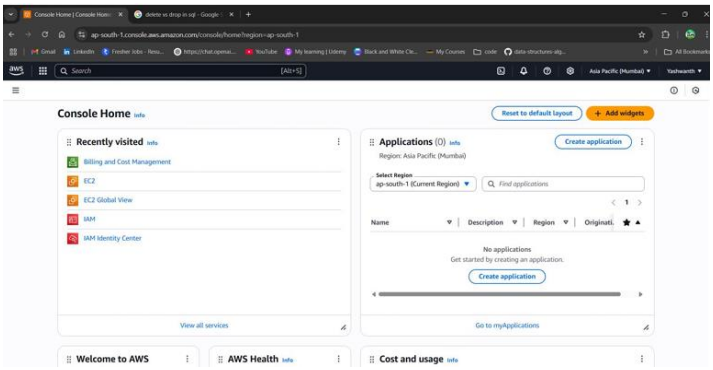
4. Security & Cleanup:

Task	Steps
Terminate EC2 Instances	- Go to **EC2 > Instances**.- Select instance → Actions → **Terminate**.
Delete Key Pairs	- Go to **EC2 > Key Pairs**.- Delete unused key pairs.
Delete Security Groups	- Delete any **custom security groups** created. *(Default cannot be deleted)*
Monitor Billing	- Go to **Billing Dashboard**:Check **Free Tier usage**, **Current month bill**.Set **Budgets & Alerts** to monitor charges.
Cost Optimization Tips	- Use **AWS Cost Explorer**.- Enable **Free Tier alerts** under Billing Preferences.

Tips & Best Practices:

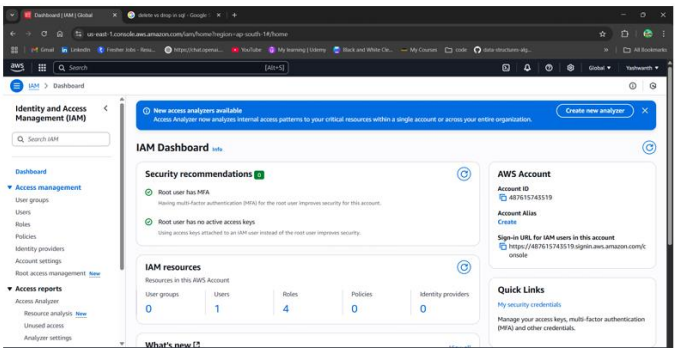
Tip	Description
Avoid Running Overnight	Always **terminate EC2 instances after testing**, especially before sleeping or logging off. Even free-tier eligible instances may incur charges if you exceed limits or add premium services (like EBS, snapshots, etc.).
Monitor Billing Dashboard	After every practice session, visit **AWS Billing Dashboard** to ensure no unexpected usage. Set up **alerts/budgets** under **Billing > Budgets** to notify you if charges start accumulating.
Explore AWS CLI (Advanced)	Once comfortable with Console, try **AWS CLI**:- Install AWS CLI locally- Configure it using IAM user credentials (`aws configure`)- Practice EC2 launch/terminate via command line.This helps in automation and DevOps workflows.

Reference :

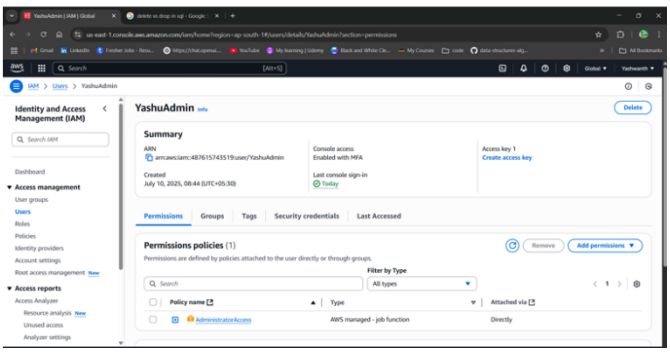


Aws homepage

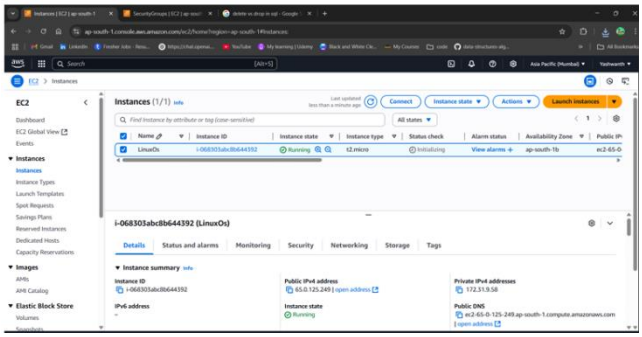
MFA:

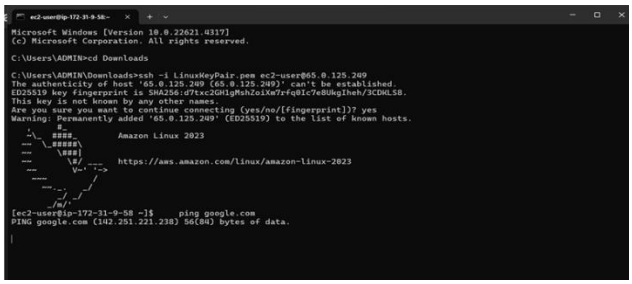


IAM USER:

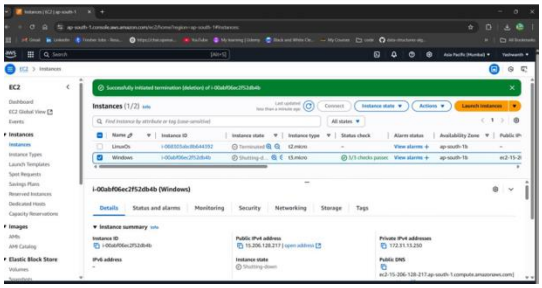
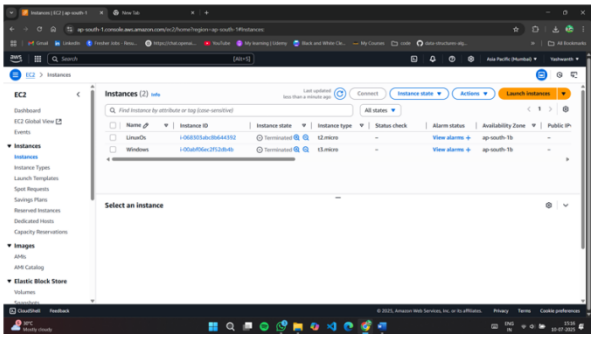


Deploying Linux (t2.micro) Instances in Mumbai Region:





Deploying Windows (t2.micro) Instances in Mumbai Region:



Terminated All Instances: